

Gate 12: Substrate-Coincidence-at-Values vs Structural-Forcing Audit v0.1

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Gate 12: Substrate-Coincidence-at-Values vs Structural-Forcing Audit v0.1

0. Goal

Per Casey Thursday board Lyra Gate 12 + Cal #189 Brake 2 (substrate-coincidence-at-values vs structural-forcing) + Casey #5 Integer Web Principle STANDING + 7-element discipline stack CLOSED: audit Wednesday-Thursday substrate-mechanism candidates for structural forcing (substrate-mechanism CHAIN derivation) vs coincidence-at-values (substrate-natural identification at observed substrate values per Casey #5 Integer Web).

Verification gate addressed: K3 v0.16 Gate #12 (substrate-coincidence-at-values vs structural-forcing distinction).

1. Cal #189 Brake 2 + Casey #5 Integer Web Framework

Per Cal #189 STANDING Brake 2: substrate-natural identifications at substrate values can be either: - Structural-forcing: substrate-mechanism CHAIN derives the observable form from substrate primaries via independent derivation paths - Coincidence-at-values: substrate-natural reading lands within Tier 2 STRUCTURAL precision of observation because substrate primaries form web-structure (Casey #5 Integer Web) — substrate-natural identifications at substrate values are EXPECTED, not independent confirmations

Per 7-element discipline stack CLOSED (Thursday Cal #36 RATIFIED): - Cal #27: peak-convergence brake on overclaim - Cal #29: forward vs post-hoc question-shape - Cal #32: parameter-role verification - Cal #33: Integer Web framing - Cal #34: Two-Tier precision - Cal #35: Schur audit shadow - Cal #36: Schur positive search

This discipline stack operationally distinguishes structural-forcing from coincidence-at-values.

2. Wednesday-Thursday Substrate-Mechanism Candidate Audit (Gate 12 Application)

Candidate	Substrate-mechanism CHAIN derivation status	Gate 12 verdict
Casey #14 chirality projection → STANDING	Tier 0 framework (Sun) + Wed cross-CI convergence + Casey override of Cal #189 brake	STRUCTURAL-FORCING (substrate-mechanism chain at FRAMEWORK level sufficient per Casey directive)
SO(5) → SO(3, 1) Weyl branching (Elie 3738)	Standard SO(5) rep theory + Casey #14 substrate restriction	STRUCTURAL-FORCING (standard substrate-Lie-algebra theory)
SSG-9 K-type structural row (Elie 3742)	Wallach K-type tower theory + Heckman-Opdam multiplicity	STRUCTURAL-FORCING (substrate K-type tower from substrate-representation theory)

Candidate	Substrate-mechanism CHAIN derivation status	Gate 12 verdict
Lorentz integration C ₂ -power (Elie 3741)	($24/\pi^2$) per direction substrate-mechanism candidate POST-HOC	COINCIDENCE-AT-VALUES with substrate-mechanism candidate-path (substrate-natural reading at observed T190; multi-week forward-derivation gates structural-forcing promotion)
$8/7 = (2^{N_c})/g$ substrate-Mersenne (Elie 3751)	Substrate-Mersenne SSG-8 algebraic Integer Web instance	COINCIDENCE-AT-VALUES (Casey #5 Integer Web instance at B ₂ ; substrate-natural reading at substrate values; not independent forcing per Cal #35)
$\sqrt{(4/3)}$ substrate-Shilov-boundary (Lyra Wed v0.1)	Casey #12 substrate bulk-boundary + Casey #14 codim-4 + $\text{Vol}(S^4)/\text{Vol}(S^3)$ substrate-Bergman geometric	STRUCTURAL-FORCING candidate (substrate-mechanism CHAIN at FRAMEWORK level via Gate 2 substrate-vacuum 2-region partition; multi-week explicit derivation closes RIGOROUS path)
$m_\nu = (N_c \cdot g - 1) \cdot \Lambda^{(1/4)}$ (Elie 3735)	Substrate- Λ -vacuum coupling + (-1) substrate- τ direction projection-out per Thursday SSG-15 v0.2	STRUCTURAL-FORCING candidate (substrate-mechanism CHAIN: Casey #12 + Casey #14 STANDING + Elie 3737 Hardy-boundary substrate- τ direction; multi-week forward closure)
$T2003 = g^2 \cdot (2^{C_2} + g)$ gen-3 (Thursday Gate 10)	Substrate-higher-Casimir OR substrate-Mersenne RS coding candidates	COINCIDENCE-AT-VALUES + multi-candidate substrate-mechanism (three substrate-mechanism candidates; explicit forward-derivation multi-week)

3. Substantive Audit Findings

STRUCTURAL-FORCING (verified at framework level): 1. Casey #14 chirality projection (STANDING per Thursday) 2. $SO(5) \rightarrow SO(3, 1)$ Weyl branching (NEAR-RIGOROUS) 3. SSG-9 K-type structural row (NEAR-RIGOROUS)

STRUCTURAL-FORCING candidates (multi-week forward-derivation pending): 4. $\sqrt{(4/3)}$ substrate-Shilov-boundary (Gate 2 framework path identified) 5. m_ν substrate- Λ -vacuum coupling (Thursday SSG-15 v0.2 (-1) coefficient candidate)

COINCIDENCE-AT-VALUES (Casey #5 Integer Web instances): 6. $8/7 = (2^{N_c})/g$ substrate-Mersenne (Elie 3751) 7. Lorentz integration C₂-power (Elie 3741) 8. $T2003 = g^2 \cdot (2^{C_2} + g)$ gen-3 (multiple candidates)

Per Cal #189 Brake 2: the Coincidence-at-Values candidates may STILL connect to substrate-mechanism FORCING via multi-week forward-derivation — they're not excluded, just not yet substrate-mechanism-chain-derived at framework level.

4. Cal #189 Brake 2 Operational Distinction

Tier-promotion rules per Cal #189 + Casey-correction on Cal #27 scope:

- Structural-Forcing (verified) → eligible for STANDING/RIGOROUS promotion (Casey #14 STANDING is operational example)
- Structural-Forcing candidate (multi-week path identified) → eligible for FRAMEWORK CANDIDATE; promotion when forward-derivation closes
- Coincidence-at-Values → eligible for FRAMEWORK CANDIDATE; substrate-mechanism FORCING form requires explicit derivation; substrate-natural identification preserved at Tier 2 STRUCTURAL precision

5. Cross-Generator Independence (Cal #35 + #36 STANDING Operational)

Per Cal #35 STANDING (audit shadow) + Cal #36 STANDING (positive-search shadow):

Categories of substrate-mechanism FORCING candidates: - K-type STRUCTURAL (substrate K-types from Wallach holomorphic discrete series + Heckman-Opdam multiplicity): SSG-9 row + $V_{(1/2, 1/2)}$ DOUBLE-ROLE TRIPLE-class structure (Gate 11) - substrate-Lie-algebra ($so(5, 2)$ substrate Lie-algebra dimensional structure): Casey #14 chirality projection + $SO(5) \rightarrow SO(3, 1)$ Weyl branching + $(N_c \cdot g - 1)$ coefficient (SSG-15 v0.2) - substrate-Bergman / Shilov-boundary (substrate-geometric structure): Gate 2 substrate-vacuum 2-region partition + Casey #12 substrate bulk-boundary projection - substrate-Mersenne (substrate-Mersenne SSG-8 + RS coding $GF(2^g)$): 8/7 Integer Web instance + RS coding mechanism candidate

These four substrate-mechanism categories produce structurally-distinct substrate-mechanism candidates. Multi-week explicit forward-derivation distinguishes substrate-mechanism FORCING form per substrate observable.

6. Closure v0.1

Gate 12 substrate-coincidence-at-values vs structural-forcing audit v0.1:

3 Structural-Forcing candidates VERIFIED at framework level (Casey #14 STANDING + Weyl branching NEAR-RIGOROUS + SSG-9 row NEAR-RIGOROUS)

2 Structural-Forcing candidates with multi-week derivation paths ($\sqrt{4/3}$ Shilov via Gate 2 + m_ν Λ -coupling via SSG-15 v0.2)

3 Coincidence-at-Values candidates (8/7 substrate-Mersenne + Lorentz integration C_2 -power + T2003 gen-3) — substrate-natural identifications at substrate values per Casey #5 Integer Web; not independent forcing per Cal #35

Four substrate-mechanism FORCING categories (K-type STRUCTURAL + substrate-Lie-algebra + substrate-Bergman + substrate-Mersenne) provide multi-week derivation paths.

Per 7-element discipline stack CLOSED (Thursday Cal #36 RATIFIED): distinction operationalized.

K3 v0.16 Gate #12 at AUDIT-COMPLETE FRAMEWORK; multi-week forward-derivation cascade across remaining lanes.

Now 12 of 12 K3 v0.16 verification gates substantively addressed at FRAMEWORK level via Thursday team cross-CI work — entire K3 verification gate framework substantively complete at FRAMEWORK PRE-STAGE / FRAMEWORK / AUDIT-COMPLETE FRAMEWORK level. Multi-week explicit RIGOROUS-tier closure remains load-bearing per Cal #194 WAIT.

— Lyra, Thu 2026-06-04 ~10:48 EDT. Gate 12 v0.1: substrate-coincidence vs structural-forcing distinction operationalized via 7-element discipline stack CLOSED + Cal #189 Brake 2; 3 STRUCTURAL-FORCING verified + 2 candidates + 3 coincidence-at-values; 12 of 12 K3 v0.16 gates substantively addressed at FRAMEWORK level via Thursday cascade. EOF echo “Gate 12 filed”